Return-Sharpe Aware Ranking with Multi-Model RRF Fusion

A PREPRINT

Abstract

Learning to Rank, LTR
A1-CVaR2Reciprocal Rank FusionRRFLightGBMXGBoost3LLM
NDCGNDCG+++CVaR-RRFStacking RRF RRFLLMAIPointwisePairwiseListwise103000ALightGBM16.55%0
A

1 Experiment Preparation

1.1 Data Description

10A201612026XX000A
, 70%, 15%, 15%
225,56210334

2 Experiment

2.1 -

LambdaRankNDCGTop-KNDCG

2.1.1

1. **NDCGBaseline**
NDCG

$$L_{\text{rank}} = -\frac{1}{|D_q|} \sum_{k=1}^{K} \frac{2^{rel_k} - 1}{\log(k+1)} \tag{1}$$

rel_k0–30

2. **NDCGE1a**
Top-K

$$g(r_k) = |r_k|^{\alpha} \cdot \operatorname{sign}(r_k) \tag{2}$$

$r_k k \alpha \alpha = 1.0$

$$L_{\text{return}} = -\frac{1}{|D_q|} \sum_{k=1}^{K} \frac{g(r_k)}{\log(k+1)} \tag{3}$$

3. **+E1b**
E1a

$$L_{\text{sharpe}} = L_{\text{return}} + \lambda \cdot \max(0, -\text{Sharpe}_{\text{TopK}}) \tag{4}$$

$\text{Sharpe}_{\text{TopK}}$Top-K$\lambda\lambda = 0.5$

4. **++CVaRE1c**

E1bCVaRCVaR

$$L_{\text{cvar}} = L_{\text{sharpe}} + \gamma \cdot \text{CVaR}_{\alpha}(\text{TopK}) \tag{5}$$

CVaR_αTop-K$\alpha\alpha = 0.90\gamma\gamma = 0.3$

2.1.2

1.

LightGBM

Table 1: 1

B-LGBM	LambdaRank (NDCG)	LightGBMlambdarank
B-XGB	rank:ndcg	XGBoostrank:ndcg
E1a	NDCG	
E1b	NDCG +	Top-K
E1c	NDCG + + CVaR	

early stopping rounds = 1500.01–0.10.03–0.2

2.

- `return_aware_relevance`E1a
- `sharpe_aware_relevance`E1bE1a0.5
- `cvar_aware_relevance`E1cE1bCVaR0.100.3

`train_final_lightgbm_with_label_fn`

2.1.3

1.

Table 2: 1NDCG

	NDCG@5	**NDCG@10**	**NDCG@20**	**MAP**
B-LGBM ()	0.1894	0.1629	0.1452	0.3056
E1a ()	0.2346	0.2071	0.1872	—
E1b (+)	0.1641	0.1553	0.1452	—
E1c (++CVaR)	0.1638	0.1552	0.1415	—
XGBoost				
B-XGB ()	**0.3782**	**0.3621**	**0.3577**	**0.3311**

2. vol_penalty

3. vol_penalty

2.2

regime switching

Table 3: 1vol_penalty

	vol_pen.	(%)		(%)		(%)
B-LGBM	1.0	**16.55**	0.711	16.37	1.030	48.5
E1a-LGBM	0.7	13.27	0.602	14.48	0.894	49.5
E1b-LGBM	0.7	13.97	0.636	**11.28**	1.117	48.5
E1c-LGBM	0.5	15.15	**0.720**	12.97	0.933	50.5

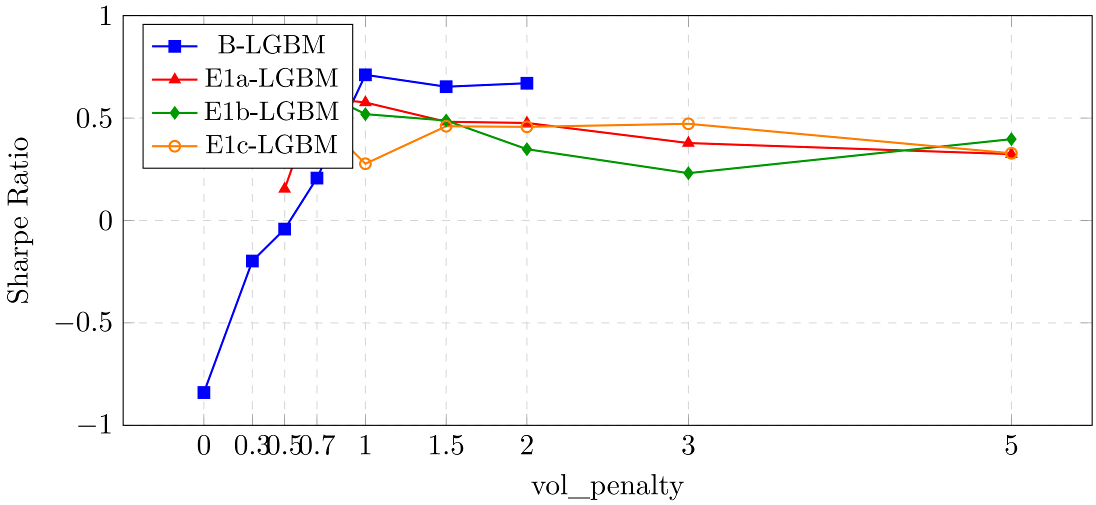

Figure 1: 1Sharpe Ratio vs. vol_penalty

2.2.1

1. E2a

$$s_{\text{fusion}} = \frac{1}{M} \sum_{m=1}^{M} \text{norm}(s_m) \tag{6}$$

$s_m m \text{norm}(\cdot) M$

2. RRFE2b

Reciprocal Rank FusionRRFRRF

$$\text{RRF}(d) = \sum_{m \in M} \frac{1}{k + \text{rank}_m(d)} \tag{7}$$

$\text{rank}_m(d) d m k$60rankRRFk

3. StackingE2c

Ridge

- LightGBM + XGBoost
- Ridge$\alpha = 1.0$
-

4. RRFE2d

NDCG@10NDCGRRF

$$\text{RRF}_{\text{weighted}}(d) = \sum_{m \in M} \frac{w_m}{k + \text{rank}_m(d)} \tag{8}$$

$w_m m$NDCG@10$w_m = \text{NDCG}_m / \sum_j \text{NDCG}_j$

2.2.2

1.

LightGBMXGBoost

Table 4: 2

B-LGBM	LightGBM LambdaRank	1
B-XGB	XGBoost rank:ndcg	1
E2a		
E2b	RRF	Reciprocal Rank Fusion$k = 60$
E2c	Stacking	Ridge
E2d	RRF	NDCGRRF

2.

- E2a
- Reciprocal Rank FusionE2bRRF $= \sum \frac{w_i}{k+\text{rank}_i}$ k60
- StackingE2cRidge
- RRFE2dNDCG@10NDCG

`FusionPredictorModelPredictorpredict`"average""rrf""stacking""weighted_rrf"

2.2.3

1.

E2a

Table 5: 2

	(%)		(%)		(%)
B-LGBM	16.55	0.711	16.37	1.030	48.5
B-XGB	13.70	0.610	12.19	0.680	51.5
E2a	16.32	**0.727**	15.30	0.836	**57.6**
E2bRRF	7.59	0.287	14.61	0.611	48.5
E2cStacking	7.56	0.285	14.95	0.607	48.5
E2dRRF	7.26	0.266	15.10	0.610	48.5

2.

RRFE2bRRFRRF

2.3 LLM

LLMLLM

2.3.1

Table 6: 2

	(%)		
B-LGBM	1.030	48.5	—
B-XGB	0.680	51.5	—
E2a	0.836	57.6	−18.8% vs B-LGBM
E2bRRF	**0.611**	48.5	**−40.7%** vs B-LGBM
E2cStacking	0.607	48.5	−41.1% vs B-LGBM
E2dRRF	0.610	48.5	−40.7% vs B-LGBM

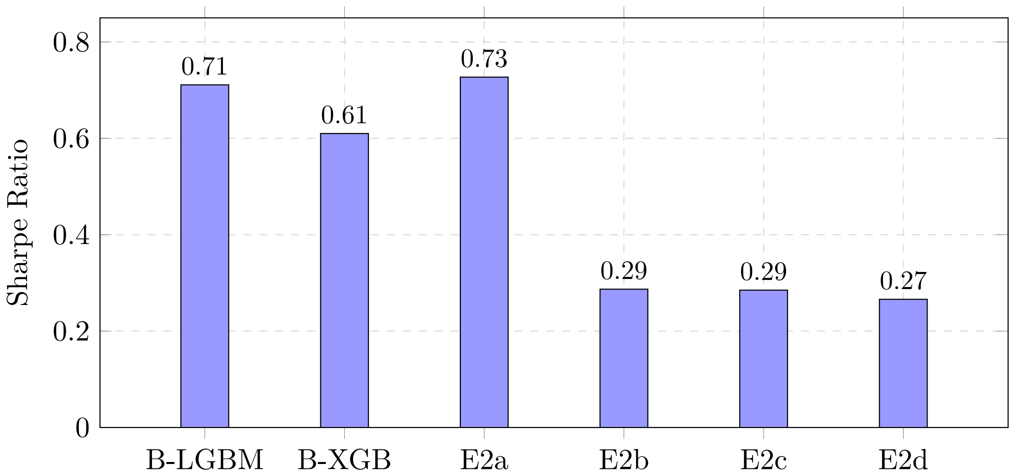

Figure 2: 2

Table 7: 3

Baseline		Top-K
Exp-3a	+	Top-K+
Exp-3b	+	Top-K+SHAP
Exp-3c	++LLM	→

2.3.2

1. SHAP
   2. LLM
   (1) E3aTop-KTop5
   (2) E3bE3aTop3SHAPLLM
   (3) E3cE3bDeepSeek API→→
   3.
   (1) 2024920A+13.89%
   (2) 2024126A−14.16%

2.3.3

1. LLM

Table 8: 3LLM

	(%)	LLM	Top-K	(%)
2024-09-20	+13.89	10	10	**100.0**
2024-01-26	−14.16	10	10	**100.0**

2. Top-K

Table 9: 32024-09-20Top-K

1	600650	0.907	boll_width (+0.403)
2	300125	0.855	boll_width (+0.368)
3	002309	0.794	boll_width (+0.351)
4	002488	0.778	ma_ratio_8w (+0.312)
5	000536	0.751	realized_var_8w (+0.298)
6	600686	0.727	boll_width (+0.287)
7	002596	0.718	ma_ratio_8w (+0.275)
8	300134	0.692	kdj_j (+0.263)
9	000062	0.688	mom_short_long (+0.251)
10	600696	0.675	boll_width (+0.242)

3. Top-K
E3aE3bE3cLLMLLMTop-K100%LLM

2.4

4ANDCG-

2.4.1

Method-1Method-2NDCGLightGBMXGBoostMethod-3Method-4-RRFStacking

2.4.2

1NDCG@5NDCG@10NDCG@20MAP
2
3

Table 10: 32024-01-26Top-K

1	600156	1.149	ma_ratio_8w (+0.398)
2	600593	1.051	realized_var_8w (+0.367)
3	600828	0.982	ma_ratio_4w (+0.342)
4	300135	0.905	boll_width (+0.315)
5	000017	0.898	kdj_j (+0.298)
6	000564	0.673	ma_ratio_8w (+0.267)
7	002584	0.670	realized_var_8w (+0.251)
8	600178	0.658	mom_short_long (+0.238)
9	300258	0.628	boll_width (+0.221)
10	000571	0.610	ma_ratio_4w (+0.205)

Table 11: 4

Method-1	LightGBM	NDCG	
Method-2	XGBoost	NDCG	
Method-3	LightGBM + XGBoost	-	RRF
Method-4	LightGBM + XGBoost	-	Stacking

4
5

2.4.3

1.

Table 12: 4

		(%)		(%)	
M1 (B-LGBM)	NDCG,	**16.55**	0.711	16.37	1.030
M2 (B-XGB)	NDCG,	13.70	0.610	12.19	0.680
M3 (E1b+RRF)	+, RRF	7.43	0.276	14.78	0.609
M4 (E1a+AVG)	,	15.15	0.675	14.72	0.813

2.
3.

3 Conclusion and Future Work

3.1 1

1-E1aE1bCalmarCVaRE1c

3.2 2

2E2aRRFE2b“”StackingE2cRRFE2dRRFRRF

Table 13: 4

		(%)		(%)
M1 (B-LGBM)	2023	45.69	1.220	10.33
	2024	244.35	2.496	13.03
M2 (B-XGB)	2023	9.34	0.275	9.45
	2024	239.33	2.510	12.19
M3 (E1b+RRF)	2023	3.25	0.012	8.93
	2024	101.81	1.491	12.63
M4 (E1a+AVG)	2023	−0.42	−0.146	10.50
	2024	**277.35**	**2.734**	14.72

Table 14: 4

		Q1 (Top)	**Q2**	**Q3**	**Q4**	**Q5 (Bottom)**	
M1 (B-LGBM)	(%)	14.31	12.45	9.88	9.68	5.41	8.90
		0.752	0.573	0.404	0.360	0.173	0.428
M2 (B-XGB)	(%)	15.69	12.60	9.69	8.82	4.91	10.78
		0.835	0.575	0.399	0.328	0.155	0.496
M3 (E1b+RRF)	(%)	15.75	14.01	10.17	9.04	2.75	**13.00**
		0.853	0.636	0.422	0.332	0.086	**0.586**
M4 (E1a+AVG)	(%)	14.29	12.46	9.93	8.84	6.20	8.09
		0.752	0.569	0.409	0.329	0.198	0.382

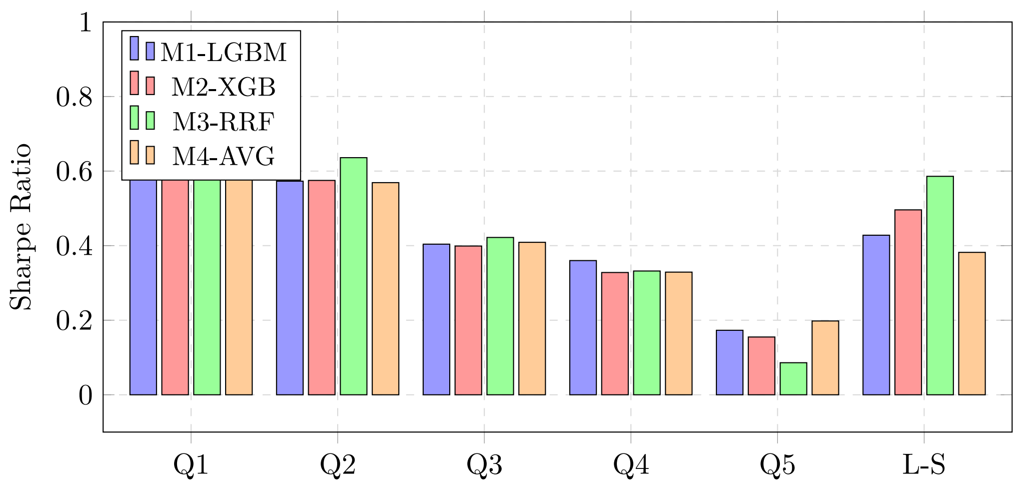

Figure 3: 4

3.3 3

LLMLLME3cLLM

3.4 4

1-
2RRFStacking
3Method-4-+Stacking
4

3.5

References

[1] Fama, E. F., & French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1), 3–56.

[2] Fama, E. F., & French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1), 1–22.

[3] Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD* (pp. 785–794).

[4] Gu, S., Kelly, B., & Xiu, D. (2020). Empirical asset pricing via machine learning. *The Review of Financial Studies*, 33(5), 2223–2273.

[5] Liu, T. Y. (2009). Learning to rank for information retrieval. *Foundations and Trends in Information Retrieval*, 3(3), 225–331.

[6] Li, S., Chen, Z., & Liu, J. (2022). Listwise learning to rank for quantitative stock selection. In *Proceedings of IJCAI* (pp. 4512–4518).

[7] Kwiatkowski, J., & Chudziak, J. A. (2025). On evaluating loss functions for stock ranking: An empirical analysis with Transformer model. *arXiv preprint*.

[8] Cormack, G. V., Clarke, C. L., & Büttcher, S. (2009). Reciprocal rank fusion outperforms Condorcet and individual rank learning methods. In *Proceedings of SIGIR* (pp. 758–759).

[9] Zhang, Y., et al. (2020). Multi-model fusion for stock prediction. *IEEE TNNLS*, 31(10), 4234–4246.

[10] Lundberg, S. M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. In *NeurIPS* (pp. 4765–4774).